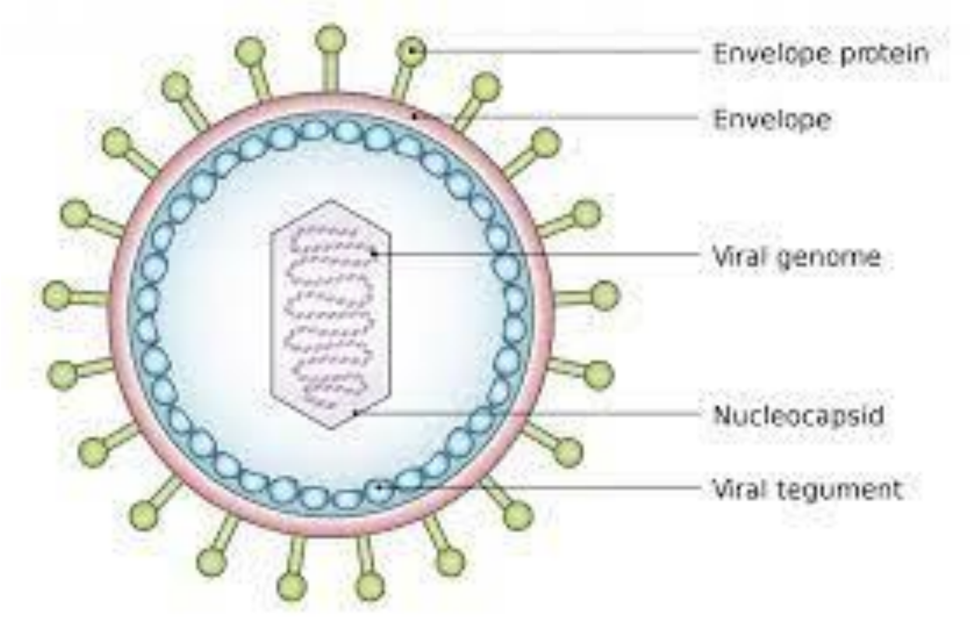
Viral infection

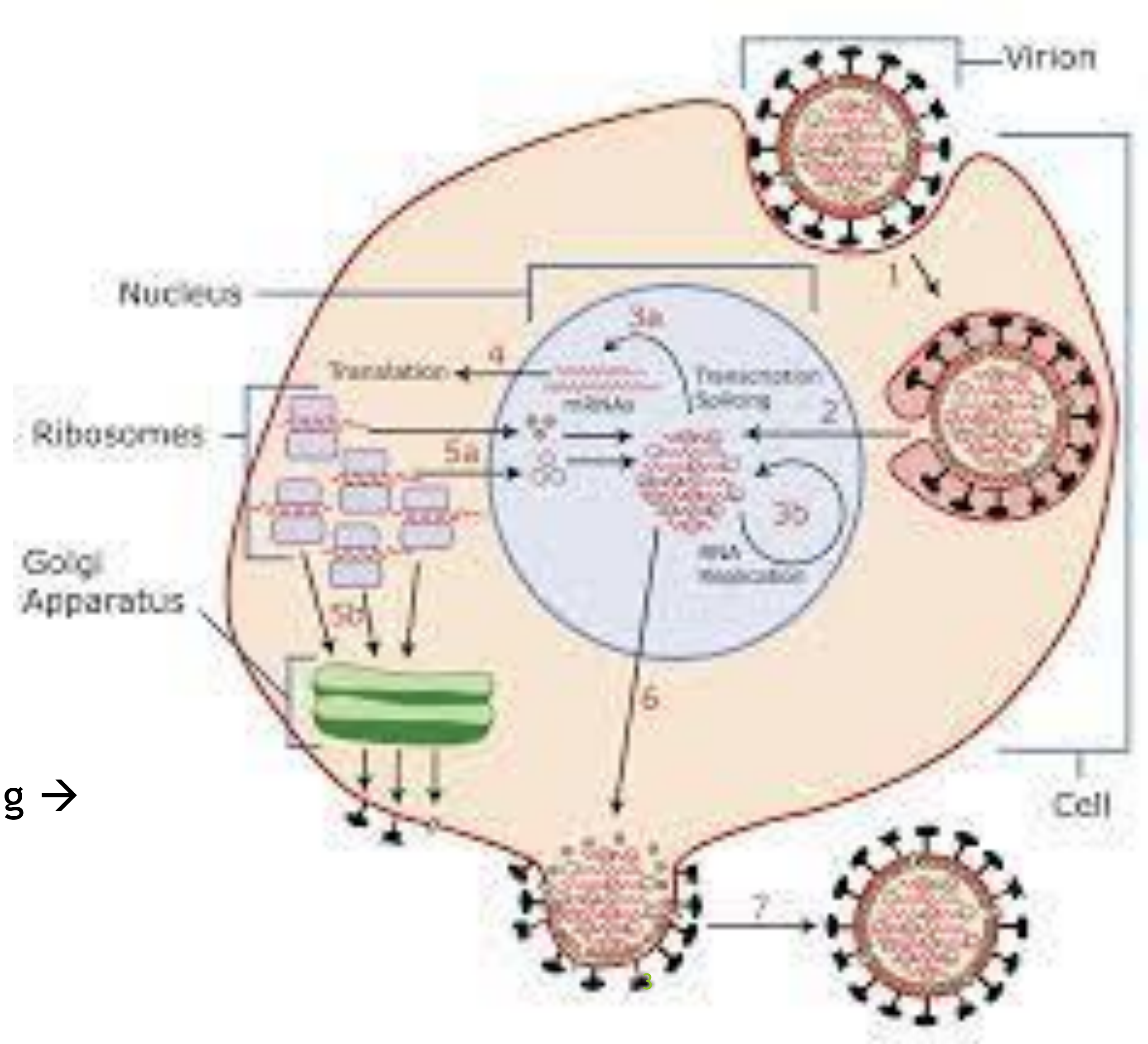
Dr emangomaa

Consultant dermatologist

Viral infection

- genom one type only
- DNA only OR RNA only.
- Opligate intra cellular
- Host specific.
- Tropisim.





Viral replication → attachment → entery → uncoating → replication & translation of viral nucleic acid → assembling → release.

Effects of viral infection on host cells

1. Lytic → HSV buallae.
2. Proliferation → HPV wart.
3. Carcinogenesis → HPV.
4. Persistant infection → continuous viral replication & normal cell.
1. Latency → no replication, varicella herpis.

Viral infection in dermis

Certain

Possible
PR

Kcs

Systemic
With
exanthem

Cutaneous
Manifestations
H.C.V.

Degeneration
H.S.V.
Varicella

Proliferation
H.P.V. → wart
MC

HHV

THE EIGHT HUMAN HERPESVIRUSES, THEIR CLASSIFICATION AND DIFFERENTIAL DIAGNOSES				
		Predominant cell type infected		
Human herpesvirus	Classification	Lytic infection	Latent infection	Differential diagnosis
Herpes simplex virus type 1 (HSV-1) (HHV-1)	Alphaherpesvirinae	Epithelial cells	Neurons	Orolabial herpes: aphthous stomatitis, erythema multiforme major, Stevens–Johnson syndrome, herpangina, pharyngitis (e.g. due to EBV), oral candidiasis, mucositis secondary to chemotherapy
Herpes simplex virus type 2 (HSV-2) (HHV-2)	Alphaherpesvirinae	Epithelial cells	Neurons	Genital herpes: trauma, chancre (primary syphilis), aphthae, EBV-related ulcers, CMV-related ulcers (AIDS), chancroid, granuloma inguinale, lymphogranuloma venereum
Varicella–zoster virus (VZV) (HHV-3)	Alphaherpesvirinae	Epithelial cells	Neurons	Varicella: vesicular viral exanthem (e.g. coxsackievirus A6), PLEVA, rickettsialpox, disseminated HSV, insect bites, drug eruption, scabies, smallpox
				Herpes zoster: zosteriform HSV, contact dermatitis, phytophotodermatitis, bullous impetigo, cellulitis
Epstein–Barr virus (EBV) (HHV-4)	Gammaherpesvirinae	B cells, epithelial cells	B cells	Group A streptococcal infection, acute viral hepatitis, drug reaction with eosinophilia and systemic symptoms (DRESS), toxoplasmosis, lymphoma; primary CMV, HHV-6 and HIV infections
				For genital ulcers: see genital herpes
Cytomegalovirus (CMV) (HHV-5)	Betaherpesvirinae	Lymphocytes, macrophages, endothelial cells	Lymphocytes, macrophages	EBV-induced infectious mononucleosis, toxoplasmosis, viral hepatitis, lymphoma
				For genital ulcers: see genital herpes
Human herpesvirus 6 (HHV-6)	Betaherpesvirinae	$CD4^+$ T cells	Lymphocytes, monocytes	Viral exanthems (e.g. measles, rubella; enterovirus, adenovirus, EBV, and parvovirus infections), scarlet fever, Rocky Mountain spotted fever, Kawasaki disease
Human herpesvirus 7 (HHV-7)	Betaherpesvirinae	T cells	T cells	Similar to HHV-6
Human herpesvirus 8 (HHV-8)	Gammaherpesvirinae	Lymphocytes	Lymphocytes, endothelial cells	Acroangiodermatitis (pseudo-KS), bacillary angiomatosis, ecchymosis, hemangioma, angiosarcoma, pyogenic granuloma, pseudolymphoma/lymphoma

Herpes simplex virus
HSV 1 & 2

- Epidemiology:- commonest infection 1\3 of world population.
- Organism:- Direct contact , droplet. HSV 1 → oral & HSV 2 → genital.
- Source:- exogenous & endogenous → auto inocluation.
- Predisposing factor:- trauma.

C.Pic
Prodroma

- Groups of vesicles on erythematous base.
- Either asymptomatic or burning sensation.
- 3-7 days after exposure.
- Scalloped border→ resolution in 2-6 weeks.

Clinical variants (10)

1)oro-labial

- → cold sore / I.P. 5 days.
- HSV 1 → vermilian border.
- Fever, sore throat & erosion with grey membrane.

Fig. 80.1 Orolabial herpes simplex virus (HSV) infections. A Primary herpes gingivostomatitis due to HSV-1 in a child. Note the coalescing lesions with scalloped borders. **B** Primary versus non-primary initial HSV-2 infection in a teenager. There are grouped vesicles on an erythematous base; note the scalloped borders. **C** Recurrent herpes labialis (cold sore, fever blister). *A, Courtesy, Julie V Schaffer, MD; B, Courtesy, Jean L Bolognia, MD.*

2) genital →Painfull erosive balanitits, vulvitis & vaginitis.

3

Fig. 80.4 Recurrent genital herpes. A–C Intact grouped vesicles and/or vesiculopustules with an erythematous base on the penis (A), medial buttock (B), and above the gluteal cleft (C). The buttocks represent a common location in women. D Healing ulcerations with scalloped borders on the penis.

Fig. 80.3 Primary genital herpes. A In addition to 1–2 mm hemorrhagic crusts, there are perifollicular vesiculopustules. B Grouped vesicles and erosions in the gluteal cleft. B, Courtesy, Kalman Watsky, MD.

) gladiatorum → contact sports, wrestLers → face , neck & medial arm.

4) Herpetic whitlow → digits → HSV 2 > 1 digits → painful swelling vesicles near cuticle.

5) HS folliculitis → HSV 1 → immunocompromised.

Fig. 80.6 Herpetic whitlow. A Coalescing vesicles and erosions on the distal finger of a child. B An edematous erythematous plaque with relatively subtle central vesicle formation on the thumb of a child. C Grouped vesicles on the toe of an adult. Herpetic whitlow is sometimes misdiagnosed as cellulitis or blistering distal dactylitis, or, depending on the distribution, paronychia. C, Courtesy, Louis A Fragola, Jr, MD.

- 6) Eczema-herpiticum → Kaposi-variccelliform eruption → defect in skin barrier → HSV 1 with rapid wide spread → monomorphic & punched out erosion.

OTHER CLINICAL PRESENTATIONS OF HERPES SIMPLEX VIRUS INFECTIONS		
Form of infection	Clinical settings and HSV type(s)	Cutaneous and/or extracutaneous findings
Predominantly mucocutaneous		
Eczema herpeticum (Kaposi varicelliform eruption)	• Affects infants/children > adults with atopic dermatitis (Fig. 80.5); risk is associated with mutations in the gene encoding filaggrin • A similar presentation can occur in patients with an impaired skin barrier due to other conditions such as burns, irritant contact dermatitis, pemphigus (foliaceus, vulgaris), Darier disease, Hailey–Hailey disease, mycosis fungoides, Sézary syndrome, ichthyoses, and rarely Grover disease and pityriasis rubra pilaris with acantholysis, as well as following ablative laser procedures or application of topical 5-fluorouracil • Usually due to HSV-1	• Rapid, widespread cutaneous dissemination of HSV infection in areas of dermatitis/skin barrier disruption • Monomorphic, discrete, 2–3 mm punched-out erosions with hemorrhagic crusts (see Fig. 80.5) are evident more often than intact vesicles • May have fevers, malaise and lymphadenopathy • Occasionally complicated by bacterial superinfections (e.g. *Staphylococcus aureus*, group A streptococci) or systemic dissemination of HSV infection • Differential diagnosis includes "eczema coxsackium" due to coxsackievirus A6 infection and streptococcal infection

Fig. 80.5 Eczema herpeticum. A Monomorphic, punched-out erosions with a scalloped border in this infant with a history of facial atopic dermatitis. B Monomorphic, small hemorrhagic crusts and erosions coalescing in the popliteal fossae, an area of pre-existing atopic dermatitis. *A, Courtesy, Julie V Schaffer, MD; B, Courtesy, Kalman Watsky, MD.*

7) Ulcerative HS → with hemorrhagic crustules → perianal & buttocks → atypical → verrucous & exophytic & pustules.

8) Ocular → HSV 2.

9)Encephalitis → HSV 1

Fig. 80.2 Recurrent herpes simplex virus type 1 infection on the cheek. Occasionally, such lesions are misdiagnosed as cellulitis or bullous impetigo. *Courtesy, Kalman Watsky, MD.*

10) Maternal neonatal congenital HSV

- Intra-uterine:- TORCH – 1st 20 weeks of gestation → abortion, stillbirth & congenital abnormalities.
- Neonatal → 1:10,000 → HSV 2 → 80 % & HSV 1 → 20%.

- Modes of transmission

-During pregnancy:- rare, disseminated infection from mother.

-intrapartum:- 90% vaginal delivery with genital herpes → 30-50 % 1ry & 1-3% 2ry

-Post-natal:- from other mother.

C.Pic of Maternal neonatal congenital HSV

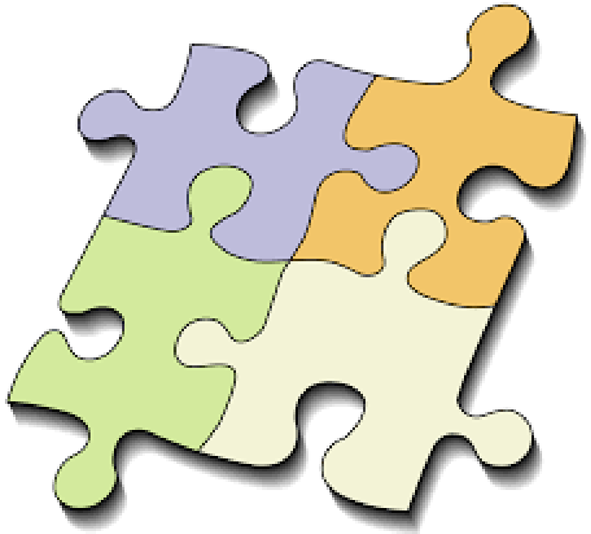
- Skin:- scalp /face→ vesicles → bullae → erosion.
- Systemic → encephalitis, hepato-adrenal necrosis & pneumonia.
- Mortality > 50% without ttt & 15% with ttt

Prevention & ttt

- Active 1ry infection → delivery →coesercan & acyclovir.
- 3rd trimester → acyclovir 36 weeks till delivery.
- History of HSV → baby monitoring.
- Neonatal HSV → IV acyclovir 60 mg/kg/day on 3 doses & oral 6mg.

Investigations of HSV

- **T-zank smear**:- fresh vesicles →scarb pus → fixed on glass slide → Giems → multinucleated syncytial giant cells → [Jigsaw puzzle]
- Electron microscopy →
- direct fleuresent AB
- PCR.



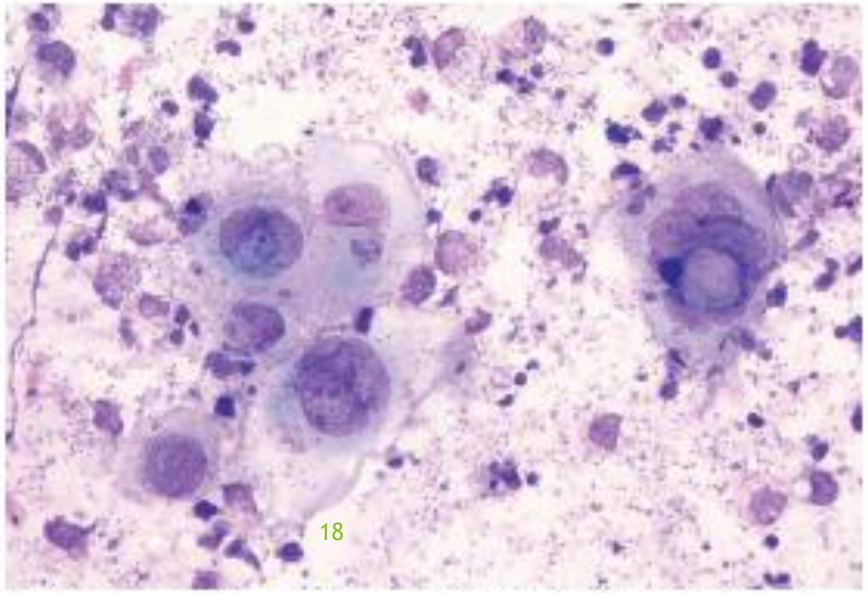


Fig. 80.11 Tzanck smear. Note the multinucleated epithelial giant cells from a patient with herpes simplex viral infection. Courtesy, Louis A Fragola, Jr, MD.

- H.P.:- intra epidermal vesicular ballooning degeneration → multinucleated giant cells → eosinophilic inclusion body.

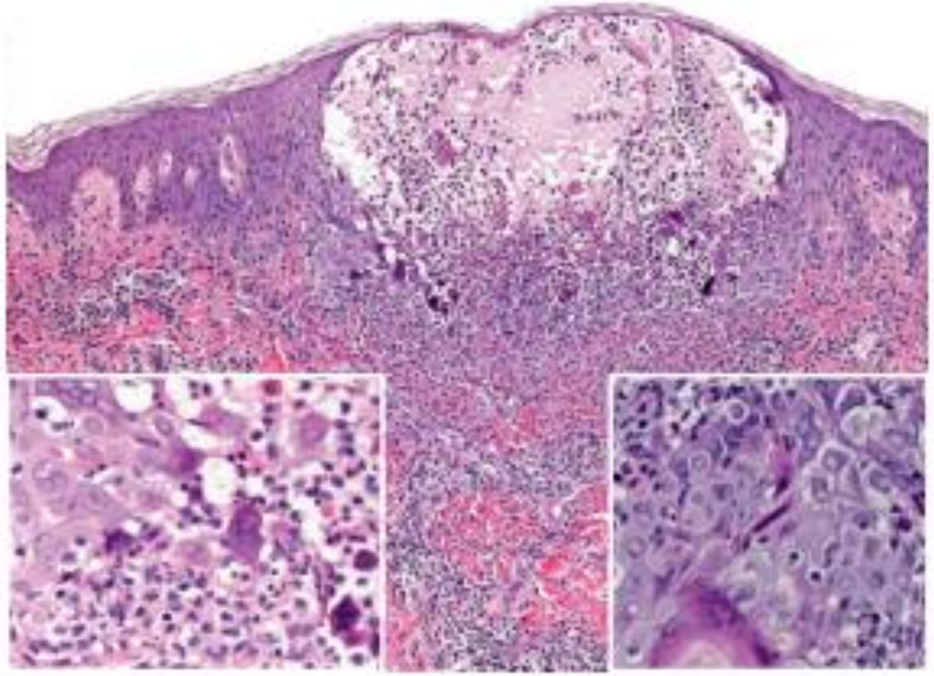


Fig. 80.12 Histology of herpes simplex viral infection. Intraepidermal vesicle with ballooning degeneration of keratinocytes and multinucleated giant cells; the latter arise from the fusion of infected keratinocytes. Note the steel-gray nuclei with margination of chromatin and inclusions (insets). Courtesy, Lorenzo

Complications

I. Cranial nerve palsies.
II. Eczema herpiticum.
III. Lymphedema.
IV. 2ry lekoderm
V. E.M.
VI. Bell’s palsy.
VII. Encephalitis (systemic spread)

DD

- Other causes of oral ulcer:- Aphthous, EMSTS, herpangina, EBV & candidiasis.
- Genital ulcer:- trauma, syphilitic chancre, chnceroid & lymphogranulo-venerum

ttt

- No ttt.
- Antiseptic.
- Antiviral→ systemic :- acyclovir 200 mg / 5 times per day

Disease context	Drug and dosage
Herpes simplex infections	
Orolabial herpes* (recurrence)	*Topical* Docosanol: 5x/d until healed Penciclovir: 1% cream applied q2h while awake × 4 days Acyclovir: 5% ointment applied q3h/6 times/day × 7–10 days Acyclovir + hydrocortisone: 5%/1% cream applied 5x/d × 5 days Acyclovir mucoadhesive tablet: 50 mg single application in the canine fossa on the side affected *Oral** Acyclovir**: 400 mg po TID × 7–10 days Famciclovir: 1.5 g po × 1 dose Valacyclovir: 2 g po BID × 1 day
Genital herpes (first episode)	Acyclovir: 200 mg po 5x/d × 10 days or 400 mg po TID × 10 days Famciclovir: 250 mg po TID × 10 days Valacyclovir: 1 g po BID × 10 days
Genital herpes (recurrence)	Acyclovir: 400 mg po TID × 5 days or 800 mg po BID × 5 days or 800 mg po TID × 2 days Famciclovir: 1 g po BID × 1 day or 500 mg po × 1 dose then 250 mg po BID × 2 days or 125 mg po BID × 5 days Valacyclovir: 500 mg po BID × 3 days or 1 g po daily × 5 days
Chronic suppression	Acyclovir: 400 mg po BID Famciclovir: 250 mg po BID Valacyclovir: 500 mg po daily for <10 outbreaks/year or 1 g po daily for ≥10 outbreaks/year
Neonatal	Acyclovir: 20 mg/kg iv q8h × 14–21 days
Immunocompromised	*Recommend use until all mucocutaneous lesions are healed* Acyclovir: 400 mg po 5x/d or 5 mg/kg (if age ≥12 y) to 10 mg/kg (if age <12 y) iv q8h Famciclovir**: 500 mg po BID Valacyclovir**: 1 g po BID
Eczema herpeticum/Kaposi varicelliform eruption**	*Recommend use for 10–14 days or (especially if immunocompromised) until all mucocutaneous lesions are healed* Acyclovir: 15 mg/kg (400 mg max) po 3–5x/d or, if severe, 5 mg/kg (if age ≥12 y) to 10 mg/kg (if age <12 y) iv q8h Famciclovir: 500 mg po BID Valacyclovir: 1 g po BID
Genital herpes, recurrent in the setting of HIV infection	*Recommend use until all mucocutaneous lesions are healed* Acyclovir**: 400 mg po TID Famciclovir: 500 mg po BID Valacyclovir**: 1 g po BID
Chronic suppression in the setting of HIV infection	Acyclovir**: 400–800 mg po BID–TID Famciclovir**: 250–500 mg po BID Valacyclovir: 500 mg po BID
Acyclovir-resistant HSV in immunocompromised patients	Foscarnet: 40 mg/kg iv q8–12 h × 2–3 weeks (or until all lesions are healed) Cidofovir: 1% cream or gel daily × 2–3 weeks or (for severe disease) 5 mg/kg iv weekly × 2 weeks then every other week (together with probenecid)
Varicella-zoster virus infections	
Varicella	Acyclovir: 20 mg/kg (800 mg max) po 4x/d × 5 days Valacyclovir†: 20 mg/kg (1 g max) po TID × 5 days
Herpes zoster	Acyclovir: 800 mg po 5x/d × 7–10 days Famciclovir: 500 mg po TID × 7 days Valacyclovir: 1 g po TID × 7 days
Immunocompromised patient or disseminated disease	Acyclovir: 10 mg/kg (500 mg/m^2) iv q8h × 7–10 days *or until cropping has ceased* (depending on the setting, consider continuing until lesions are healed)

Varicella [chicken pox]

- 1ry infection with VZV.
- Epidemiology:- 90% children < 10 ys (pre-vaccum era)
- Winter spring.
- IP:- 11-20 days. / causative: VZV transmitted by airborne droplets &
- direct contact with the vesicles → extremely contagious.
- 1-2 days before vesicles – 5 days after rush (crustation).

C.pic:- all symptoms increase with age

- Prodroma:- 1-2 days.
- Skin :- 3-5 crops (polymorphism) → macule → papule → clear vesicles → red halo (dew drop on a rose petal).
- Site:- scalp, face, trunk, extrimities → sparing of distal lower ex.

Clinical variants:-

Hemorrhagic varicella.
Chronic hyper-keratotic.
Maternal.

- Investigations:-
- H.P./T.zank/microscopic & PCR.

- Complications:-
- CNS, lung → adult.
- 2ry infection & Reye's Syndrome.
- Course:-

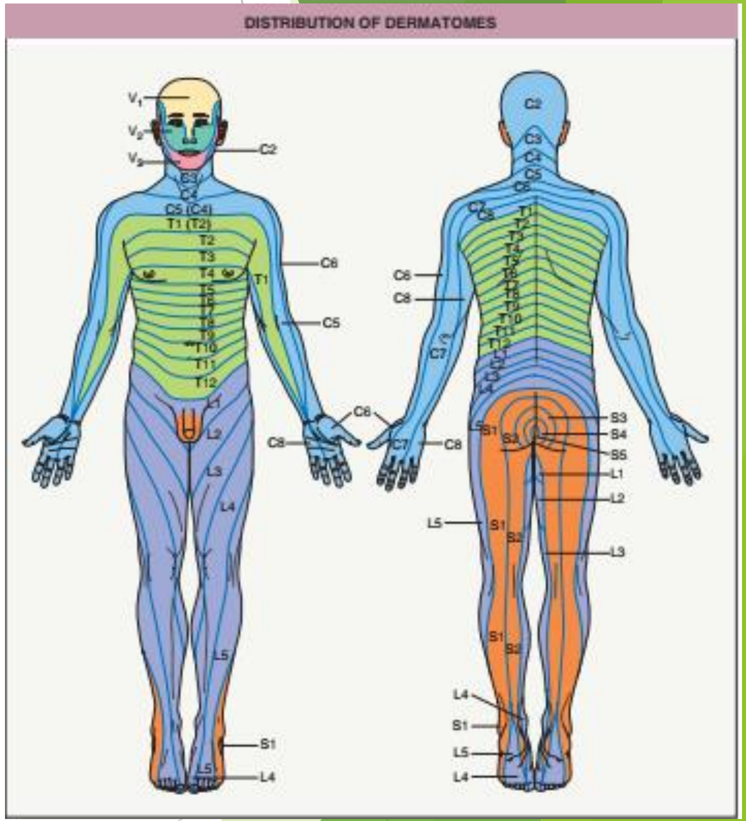
Latency → immune suppression → H.Z.

- TTT:-

Symptomatic →
antiviral.

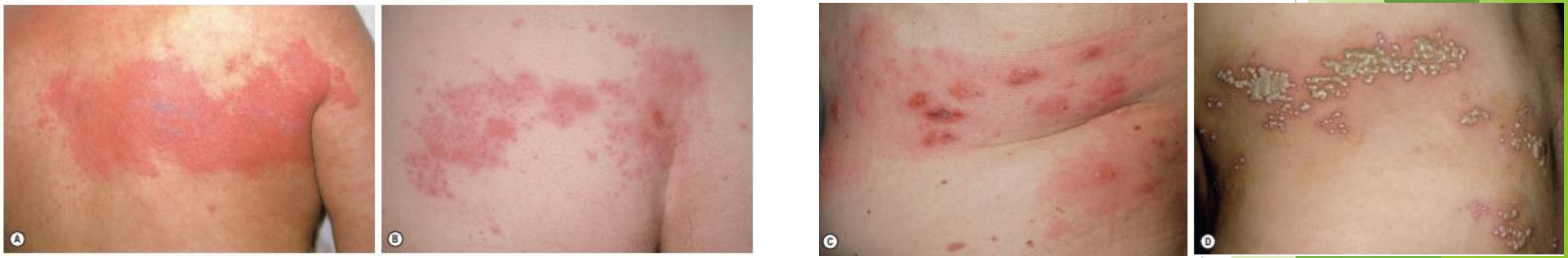
Herpes zoster

- Segmental eruption due to reactivation of latent zoster.
- Epidemiology:- Old age.
- Aetiopathogenesis:- reactivated HZV → after clinical or subclinical varicella.



C.pic

- Prodroma:- pain, pruritis without eruption → erythema → eruption or zoster sincheriete.
- Skin eruption:- grouped vesicles on erythematous base intra-dermal →thoracic, cervical & trigeminal.



Clinical variants

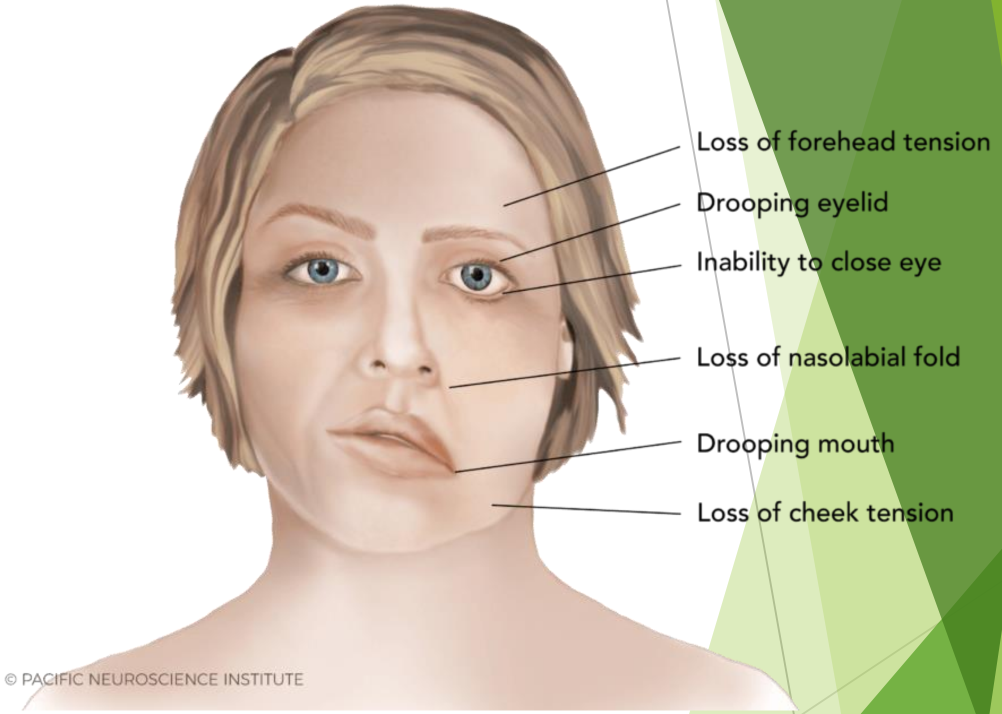
- Trigeminal

- Ophthalmic 2/3 of cases.
- Naso-ciliary nerve → Hutchinson sign.
- Ciliary ganglion →Argyll – Robertson sign.
- Pupillary constriction on accommodation not in light exposure.
- Ocular complications:- ocular muscle palsies, uveitis & keratitis.

Fig. 80.16 Facial herpes zoster. A Clustered vesicles in a linear array within the distribution of V_1. **B** In the distribution of V_3, grouped pustules on the left side of the chin and hemorrhagic crusting of the lower lip, with one lesion extending past the midline. This could be mistaken for an acneiform eruption or impetigo. **C** In the same patient, erosions are present on the left side of the tongue. The anterior two-thirds of the tongue is innervated by the facial nerve (VII; taste) as well as V_3 (sensory); see Table 80.6 for details of Ramsay Hunt syndrome. **D** Focal hemorrhagic crusting with scalloped borders arising in an erythematous patch on the forehead within the V_1 distribution. Note the periorbital edema. *Courtesy, Kalman Watsky, MD.*

❑ Facial

- HZ oticus.
  - Sensory → pain & vesicles.
- Ramsay-hunt sign.
- Geniculate ganglion.
- Eac, ear pain.
- Anterior 2/3 of the tongue & tympanic membrane.



❑ Motor

Old , malignant cranial nerve.

Ophthalmic → ophthalmoplegia.

Facial→ facial palsy.

Anogenital →defective urination & defecation.

Cervical→peripheral motor neuropathy in arm → weakness.

Thoracic→abdominal weakness & hernia.

Lumbar → motor neuropathy of LL → weakness.

CLINICAL FINDINGS RELATED TO THE DERMATOME INVOLVED IN HERPES ZOSTER		
Dermatome/nerve involved	**Cutaneous findings**	**Extracutaneous findings**
Ophthalmic division of the trigeminal nerve (V_1; ~10–15% of zoster patients)	*Hutchinson's sign* refers to increased risk of ocular involvement if skin lesions are in the distribution of the nasociliary branch, which supplies the nasal tip, dorsum and root of the nose, and medial canthus as well as the cornea	• ~50% of patients have ocular involvement, which can include conjunctivitis, (epi)scleritis, keratitis, uveitis, acute retinal necrosis, and optic neuritis • May lead to ocular scarring and visual loss
Geniculate ganglion of the facial nerve – *Ramsay Hunt syndrome*	Lesions affecting the external auditory canal, tympanic membrane, anterior two-thirds of the tongue, and/or hard palate	• Ear pain • Facial nerve paralysis (peripheral) • Loss of taste in anterior two-thirds of the tongue • Dry mouth and eyes • If the vestibulocochlear nerve is also affected: tinnitus, hearing loss and/or vertigo
Cervical dermatome	Lesions in cervical dermatome	• Peripheral motor neuropathy of the arm, which may result in muscle atrophy with gradual recovery of strength • Diaphragmatic weakness (C3–5)
Thoracic dermatome	Lesions in thoracic dermatome	• Abdominal muscle weakness • Abdominal wall pseudohernia
Lumbar dermatome	Lesions in lumbar dermatome	• Peripheral motor neuropathy of the leg, which may result in muscle atrophy with gradual recovery of strength

- Dissiminated

20 vesicles outside area of 1ry lesion.

- Maternal

Complications

1.Post-herpetic neuralgia 10-20%.

- Persistence & recurrence of pain.
- Risk factors:- old-prolonged zoster → trigeminal N. → burning pain, shooting & allodynia.

2. Cutaneous (3 comp).

CUTANEOUS DISORDERS THAT CAN DEVELOP WITHIN SITES OF HERPES ZOSTER

Acneiform **–** comedones, furunculosis, "acneiform eruption", rosacea (trigeminal distribution)

Dermatitis and urticaria **–** contact dermatitis (allergic and irritant), chronic localized urticaria

Granulomatous **–** granuloma annulare (including perforating variant), granulomatous dermatitis (sarcoidal, tuberculoid, unclassified, periadnexal, perineural; see Fig. 80.17), granulomatous vasculitis, lichenoid and granulomatous dermatitis*

Infiltrative **–** pseudolymphoma†, xanthomatous, Rosai–Dorfman disease

Malignancies, hematologic **–** leukemia (especially CLL), lymphoma cutis**

Malignancies, other **–** Kaposi sarcoma (HIV-associated form), angiosarcoma, solid-organ metastases

Papulosquamous **–** psoriasis, lichen planus, lichenoid GVHD*

Sclerosing **–** lichen sclerosus, morphea, hypertrophic scars/keloids

Ulcerative **–** trophic syndrome (trigeminal, cervical)

Miscellaneous **–** acquired perforating disorder (acquired perforating collagenosis), nodular solar degeneration, tinea, bullous pemphigoid

TTT

- Systemic antiviral→ as early as possible.
- Ttt of PHN → potent analgesic & Pain killer → Gaptin & pregabalin→ nerve block

EBV-ASSOCIATED DISORDERS
Infectious mononucleosis
Gianotti–Crosti syndrome
Genital ulcers*
Periodontal disease
Oral hairy leukoplakia
Hydroa vacciniforme and related conditions
• Typical hydroa vacciniforme
• Severe hydroa vacciniforme-like eruptions**
• Hydroa vacciniforme-like lymphoproliferative disorder***
Severe hypersensitivity to mosquito bites*
Lymphadenitis, including Kikuchi disease
Lymphoproliferative disorders and lymphomas
• Post-transplant lymphoproliferative disorder/lymphoma
• Methotrexate-associated lymphoproliferative disorder (usually in patients with rheumatoid arthritis)
• Systemic EBV+ T-cell lymphoma of childhood
• EBV+ mucocutaneous ulcer
• X-linked lymphoproliferative syndrome
• Hemophagocytic lymphohistiocytosis (susceptibility to develop)
• Lymphomatoid granulomatosis
• Burkitt lymphoma, endemic ≫ sporadic
• Extranodal NK/T-cell lymphoma, nasal type
• EBV+ diffuse large B-cell lymphoma, NOS
• Angioimmunoblastic T-cell lymphoma
• Other non-Hodgkin lymphomas (including T-cell), particularly in HIV-infected patients
• Hodgkin disease
Other malignancies
• Nasopharyngeal carcinoma
• Gastric carcinoma
• Leiomyosarcoma
Smooth muscle tumors

Fig. 80.21 Vulvar ulcers associated with primary EBV infection. EBV-related genital ulcers, which are most common in adolescent girls, are often misdiagnosed as genital herpes simplex infection and in some patients represent a variant of aphthosis (see Table 80.8).

Fig. 80.22 Ampicillin-induced eruption in a patient with infectious mononucleosis due to EBV infection. Erythematous macules and papules have become confluent on the upper

CUTANEOUS MANIFESTATIONS OF CMV INFECTIONS
Immunocompetent host
Mononucleosis-like syndrome
• Maculopapular/morbilliform eruption • Petechiae and purpura • Ampicillin-induced eruption • Urticaria • Erythema nodosum
Congenital infections
• "Blueberry muffin" lesions (extramedullary hematopoiesis; see Ch. 121) • Petechiae and purpura • Vesicles
Immunocompromised host
Cutaneous vasculitis Maculopapular/morbilliform eruption Petechiae and purpura Ulcers Vesicles Verrucous plaques Nodules and hyperpigmented plaques

Fig. 80.23 TORCH syndrome due to cytomegalovirus. Multiple purpuric papules of dermal hematopoiesis. *Courtesy, Mary S Stone, MD.*

TREATMENT FOR CMV INFECTIONS IN IMMUNOCOMPROMISED PATIENTS		
Medication	Route	Major side effects
Ganciclovir	iv, intravitreal implant or injection	Bone marrow suppression, GI symptoms > renal insufficiency, neuropathy
Valganciclovir	po	
Cidofovir	iv*, intravitreal injection	Renal toxicity
Foscarnet	iv, intravitreal injection	Renal toxicity
Fomivirsen	Intravitreal injection	

HHV 8 →roseola infantum

WHO CLINICAL STAGING OF HIV/AIDS IN THE SETTING OF CONFIRMED HIV INFECTION		
All ages	Adults and adolescents	Children
Clinical stage 1 – asymptomatic; immunologic correlate: >500 CD4⁺ cells/mm³*		
• Asymptomatic • Persistent generalized lymphadenopathy		
Clinical stage 2 – mild; immunologic correlate: 350–499 CD4⁺ cells/mm³*		
• Herpes zoster • Fungal nail infection • Pruritic papular eruptions • Angular cheilitis • Recurrent oral ulcerations • Recurrent/chronic upper respiratory tract infections (sinusitis, otitis media/otorrhea, tonsillitis, pharyngitis)	• Seborrheic dermatitis • Moderate unexplained weight loss (<10% of body weight)	• Extensive warts • Extensive molluscum contagiosum • Lineal gingival erythema • Unexplained persistent parotid enlargement • Unexplained persistent hepatosplenomegaly
Clinical stage 3 – advanced; immunologic correlate: 200–349 CD4⁺ cells/mm³*		
• Persistent oral candidiasis (after first 6–8 weeks of life) • Oral hairy leukoplakia • Acute necrotizing ulcerative stomatitis, gingivitis, or periodontitis • Unexplained chronic diarrhea (>1 month in adults, >2 weeks in children) • Unexplained persistent fever (≥37.6°C intermittent or constant, for >1 month) • Pulmonary tuberculosis (current) • Unexplained anemia (<8 g/dL), neutropenia (<0.5 × 10^9/L) or chronic thrombocytopenia (<50 × 10^9/L)	• Unexplained severe weight loss (>10% of body weight) • Severe bacterial infections such as pneumonia, empyema, pyomyositis, bone or joint infection, meningitis, bacteremia	• Unexplained moderate malnutrition or wasting not adequately responding to standard therapy • Recurrent severe bacterial pneumonia • Symptomatic lymphoid interstitial pneumonitis • Lymph node tuberculosis • Chronic HIV-associated lung disease including bronchiectasis
Clinical stage 4 – severe/AIDS-defining conditions; immunologic correlate: <200 CD4⁺ cells/mm³*		
• Chronic herpes simplex infection (orolabial, genital, or anorectal of >1 month's duration or visceral at any site) • Kaposi sarcoma • Extrapulmonary cryptococcosis including meningitis • Disseminated endemic mycosis (coccidioidomycosis or histoplasmosis) • Disseminated non-tuberculous mycobacterial infection • Unexplained severe wasting, stunting or severe malnutrition not responding to standard therapy (HIV wasting syndrome) • Cytomegalovirus infection (retinitis or infection of other organs; onset at age >1 month) • HIV encephalopathy • Progressive multifocal leukoencephalopathy • Extrapulmonary tuberculosis • Candidiasis of the esophagus, trachea, bronchi, or lungs • Pneumocystis pneumonia • Central nervous system toxoplasmosis (onset at age >1 month)	• Atypical disseminated leishmaniasis • Recurrent severe bacterial pneumonia • Recurrent non-typhoidal *Salmonella* bacteremia • Invasive cervical carcinoma	• Recurrent severe bacterial infections such as empyema, pyomyositis, bone or joint infection, or meningitis (but excluding pneumonia)

CLASSIFICATION OF CUTANEOUS LESIONS IN HIV

INFECTIOUS	NEOPLASTIC	OTHERS
• Herpes zoster	• Kaposi's sarcoma	• Pruritic papular eruption
• Herpes simplex	• Lymphoma	• Eosinophilic folliculitis
• Cryptococcus	• Squamous cell carcinoma	• Seborrheic dermatitis
• Histoplasmosis		• Drug eruption
• Human papillomavirus		• Vasculitis
• Impetigo		• Psoriasis
• Lymphogranuloma venereum		• Hyperpigmentation
• Molluscum contagiosum		• Photodermatitis
• Syphilis		• Atopic Dermatitis
• Furunculosis		• Hair changes
• Folliculitis		
• Pyomyositis		
• Superficial fungal infections		

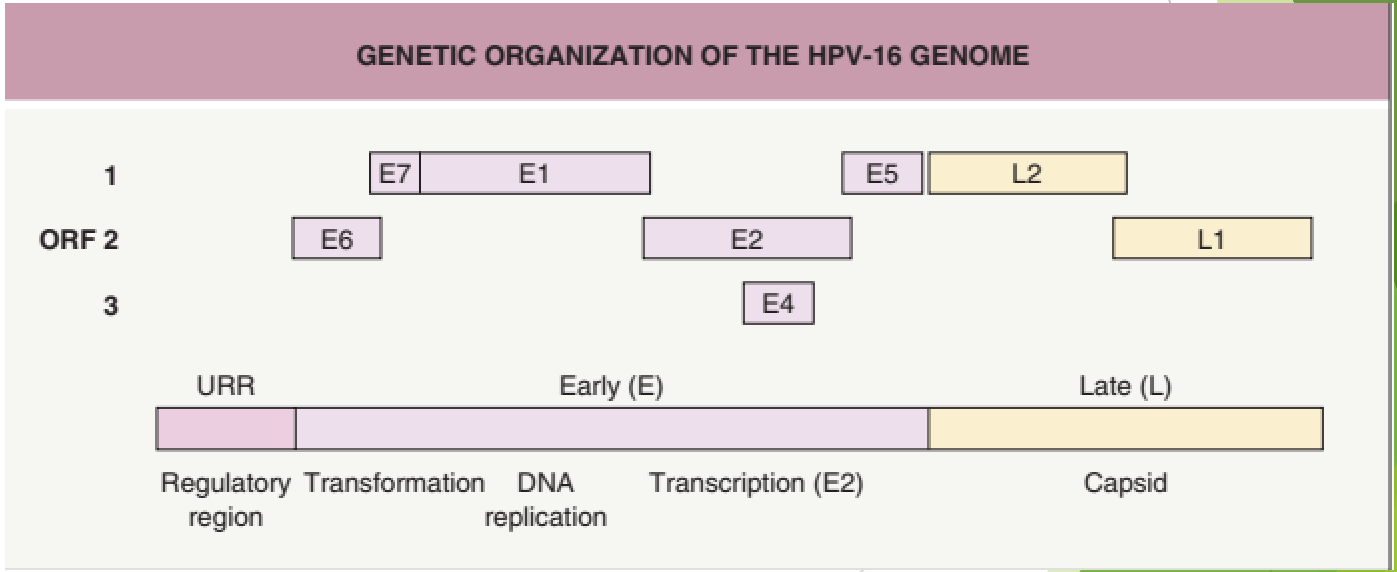
CORRELATION OF CD4$^+$ CELL COUNT WITH SPECIFIC HIV-ASSOCIATED DISORDERS				
System*	>500 CD4$^+$ cells/mm^3	<500 CD4$^+$ cells/mm^3	<250 CD4$^+$ cells/mm^3	<50 CD4$^+$ cells/mm^3
Dermatologic	Acute retroviral syndrome Oral hairy leukoplakia Vaginal candidiasis Seborrheic dermatitis	Oropharyngeal candidiasis (thrush) Herpes zoster Psoriasis, severe or refractory Eruptive atypical melanocytic nevi and melanoma Kaposi sarcoma	Eosinophilic folliculitis Seborrheic dermatitis, refractory Mollusca, extensive Bacillary angiomatosis Miliary/extrapulmonary tuberculosis Herpes simplex virus infection, disseminated Cryptococcosis, disseminated Histoplasmosis, disseminated Coccidioidomycosis, disseminated Botryomycosis Non-Hodgkin lymphoma	Large, non-healing mucocutaneous herpes simplex virus infections (e.g. perianal) Papular pruritic eruption Giant mollusca Perianal ulcers due to cytomegalovirus Aspergillosis Acquired ichthyosis *Mycobacterium avium* complex infections Major aphthae
Respiratory	Bacterial pneumonia and sinusitis	Pneumococcal pneumonia Pulmonary tuberculosis Lymphocytic interstitial pneumonia (pediatric)	*Pneumocystis jiroveci* pneumonia (PCP)	*Pseudomonas* spp. pneumonia
Nervous	Aseptic meningitis Guillain–Barré syndrome	Mononeuritis multiplex	HIV-associated dementia Cerebral toxoplasmosis Peripheral neuropathy Progressive multifocal leukoencephalopathy	Primary CNS lymphoma
Hematologic	Persistent generalized lymphadenopathy	Anemia Idiopathic thrombocytopenic purpura	Non-Hodgkin lymphoma	
Other	Myopathy	Cryptosporidiosis Cervical/anal intraepithelial neoplasia Cervical cancer Anal cancer	Esophageal candidiasis Wasting Microsporidiosis Vacuolar myelopathy Cardiomyopathy	Cryptosporidiosis, refractory Cytomegalovirus retinitis, extraocular systemic cytomegalovirus infection

CUTANEOUS SIDE EFFECTS OF ANTIRETROVIRAL THERAPY (ART)						
Type of skin reaction	NRTIs/NtRTIs	NNRTIs	Protease inhibitors	Integrase inhibitors	CCR5 inhibitors	Fusion inhibitors
Injection site reaction						
Morbilliform eruption	Emtricitabine* Abacavir**	Nevirapine > etravirine, efavirenz	Atazanavir, fosamprenavir^ > darunavir^, tipranavir^ > lopinavir			
DRESS (DIHS)	Abacavir**	Etravirine, efavirenz Nevirapine	Atazanavir, darunavir^, fosamprenavir^, tipranavir^, lopinavir	Rare	Rare	Rare
SJS/TEN	Rare	Rare Nevirapine > efavirenz, etravirine	Rare	Rare		
Lipodystrophy (also insulin resistance and elevated TTGs)	d4T, ddI	Efavirenz				
Paronychia, excessive periungual granulation tissue	Lamivudine		Indinavir			
Xerosis			Indinavir			
Hyperpigmentation of skin and nails	AZT, emtricitabine (palmoplantar)					

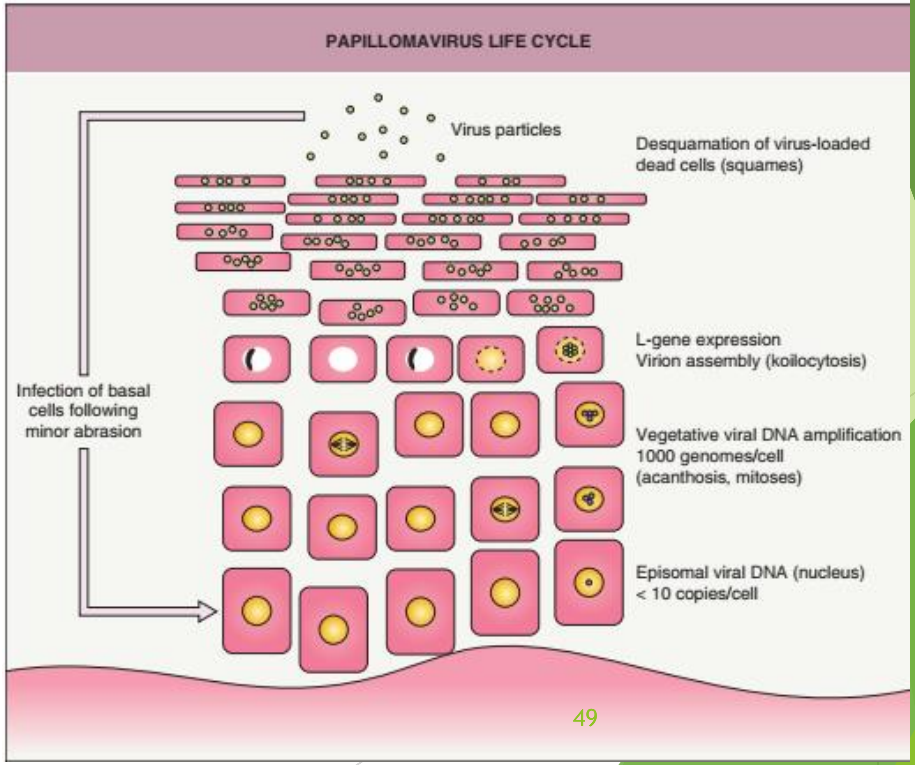
Human papilloma virus

- 150 type
- DSDNA
- No envelop
- Genom :2 region:

- Early :
  - E6,7 :oncogens stimulat proliferation , prevent apoptosis
- Late :
  - L:for capsid protein
  - L1 :major capsid protein



- Epitheliotropic :skin , mucous membrane
- Life cycle :
  - Trauma →enter virus →basal cell layer →as perminant layer → autonomus replication



Oncogenic potential of HPV

- Low risk : HPV 6-11
- High risk mucosa : 16 – 18
- High risk skin : 5-8

CLINICAL MANIFESTATIONS AND ASSOCIATED HUMAN PAPILLOMAVIRUS (HPV) TYPES		
	Frequently detected	Less frequently detected
Skin lesions		
• Common, palmar, plantar, myrmecial and mosaic warts	1, 2, 27, 57	4, 29, 41, 60, 63, 65
• Flat warts	3, 10	28, 29
• Butcher's warts	7	1, 2, 3, 4, 10, 28
• Digital squamous cell carcinoma and Bowen disease	16	26, 31, 33, 34, 35, 51, 52, 56, 73
• Epidermodysplasia verruciformis (EV)	3, 5, 8	9, 12, 14, 15, 17, 19–25, 36–38, 47, 49, 50, etc.
• EV – squamous cell carcinoma	5, 8	14, 17, 20, 47
Mucosal lesions		
• Condylomata acuminata	6, 11	40, 42–44, 54, 61, 70, 72, 81
• High-grade intraepithelial neoplasias (including cervical condylomata plana, bowenoid papulosis, erythroplasia of Queyrat) and invasive cancer	16	18, 26*, 31, 33, 35, 39, 45, 51, 52, 53*, 56, 58, 59, 62, 66*, 68, 73, 82
• Buschke–Löwenstein tumor	6, 11	
• Recurrent respiratory papillomatosis, conjunctival papillomas	6, 11	
• Heck disease (focal epithelial hyperplasia)	13, 32	

wart

epidermis

Bengin
Wart
Hpv
6-11

Malignant
Scc
EV
5-8

Anogenital

Benign
Condyloma
acummenata
6-11

Malignant
Cervical
cancer
16-18

Cutaneous wart

- Benign tumor of skin → infection by HPV
- 20-30 % of school children
- Midof infection :
  - Direct
  - Indirect